LECTURE 14 · DISCRETE MATHEMATICS

Repair relations. Discover classes.

Which missing connections are forced?



Reflexive

Every element is related to itself.

$$\forall a \in A, \quad aRa.$$

Example

$$R = \{(1,1), (2,2), (3,3)\}$$

All three required loops are present.

Non-example

$$R = \{(1,1), (2,2)\}$$

$(3,3)$ is missing.

In both examples, $A = \{1,2,3\}$ and $R \subseteq A \times A$.

Irreflexive

No element is related to itself.

$$\forall a \in A, \quad \neg(aRa).$$

Example

$$R = \{(1,2)\}$$

There are no loops.

Non-example

$$R = \{(1,1),(1,2)\}$$

The loop $(1,1)$ violates the rule.

In both examples, $A = \{1,2,3\}$ and $R \subseteq A \times A$.

Symmetric

Whenever a pair is present, its reverse is present too.

$$\forall a, b \in A, \quad aRb \Rightarrow bRa.$$

Example

$$R = \{(1,2), (2,1)\}$$

Each pair has its reverse.

Non-example

$$R = \{(1,2)\}$$

$(1,2)$ is present but $(2,1)$ is missing.

In both examples, $A = \{1,2,3\}$ and $R \subseteq A \times A$.

Antisymmetric

Two distinct elements cannot be related in both directions.

$$\forall a, b \in A, \quad (aRb \land bRa) \Rightarrow a = b.$$

Example

$$R = \{(1,1), (1,2)\}$$

The loop is allowed; no distinct pair goes both ways.

Non-example

$$R = \{(1,2), (2,1)\}$$

Both directions are present, but $1 \neq 2$.

In both examples, $A = \{1, 2, 3\}$ and $R \subseteq A \times A$.

Asymmetric

Whenever a pair is present, its reverse must be absent.

$$\forall a, b \in A, \quad aRb \Rightarrow \neg(bRa).$$

Example

$$R = \{(1, 2)\}$$

The reverse pair $(2, 1)$ is absent.

Non-example

$$R = \{(1, 1)\}$$

A loop is its own reverse, so it violates the rule.

In both examples, $A = \{1, 2, 3\}$ and $R \subseteq A \times A$.

Transitive

Every two-step chain has a direct pair joining its ends.

$$\forall a, b, c \in A, \quad (aRb \land bRc) \Rightarrow aRc.$$

Example

$$R = \{(1,2), (2,3), (1,3)\}$$

The chain $1R2$, $2R3$ has the required pair $1R3$.

Non-example

$$R = \{(1,2), (2,3)\}$$

$1R2$ and $2R3$, but $(1,3)$ is missing.

In both examples, $A = \{1,2,3\}$ and $R \subseteq A \times A$.

MINDS BEHIND THE MATHEMATICS

René Descartes

1596–1650 · France

He made equations draw pictures.

Original AI-illustrated abstract digital artwork

DESCARTES · THE LEGEND

A fly on the ceiling?

A famous later story says a fly suggested locating points by distances from walls.

Charming

An instant picture of coordinates

Caution

A legend, not secure evidence

Useful stories still deserve honest labels.

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DESCARTES · THE IDEA

Geometry becomes an equation

A point becomes an ordered pair; a curve becomes a relation.

$$x^2 + y^2 = r^2$$

Algebra can move, intersect, and transform a shape.

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DESCARTES · THE LEGACY

One bridge, two languages

Coordinates let us move between symbolic and visual reasoning.

In mathematics

Graph equations

In computing

Place pixels, objects, and data

When two representations meet, new questions become possible.

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René Descartes

EXERCISE 1

Which rules are still broken?

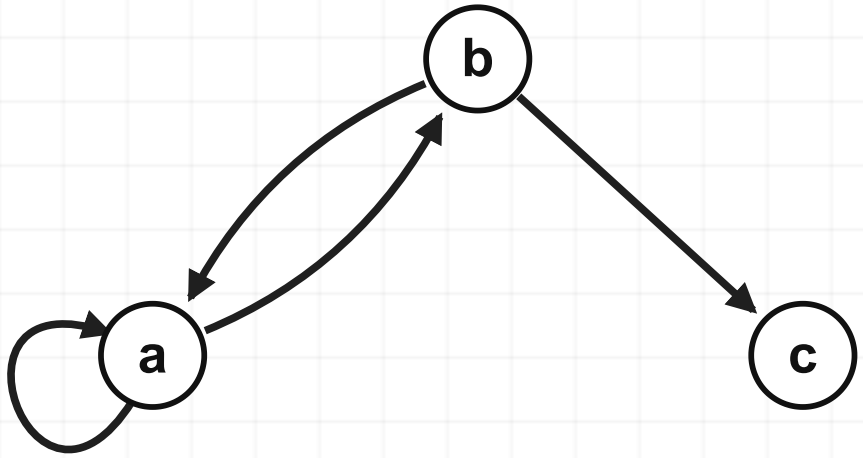
$$A = \{a, b, c\}, \quad R = \{(a, a), (a, b), (b, a), (b, c)\}.$$

01 **Reflexive? Symmetric?**

No: (c, c) is missing. No: (c, b) is missing.

02 **Transitive?**

No: aRb and bRc, but not aRc.



A closure adds only forced pairs

For reflexivity, symmetry, or transitivity, the closure is the smallest relation containing R with the chosen property.

> **Keep every original pair**
>
> $R \subseteq C$

> **Make the rule hold**
>
> **Add every pair required by the rule.**

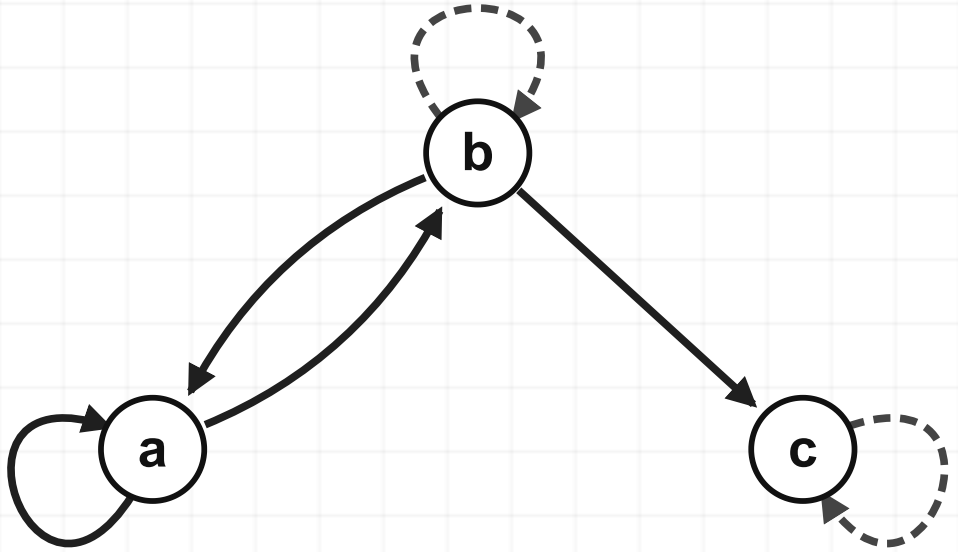
> **Add nothing unnecessary**
>
> **If $R \subseteq S$ and S has the property, then $C \subseteq S$.**

“Smallest” means contained in every relation that meets the requirements.

Reflexive closure: add the missing loops

$A = \{a, b, c\}$. Keep the four pairs of R.

01 **Which pairs must be added?**
(b, b) **and** (c, c)**.**

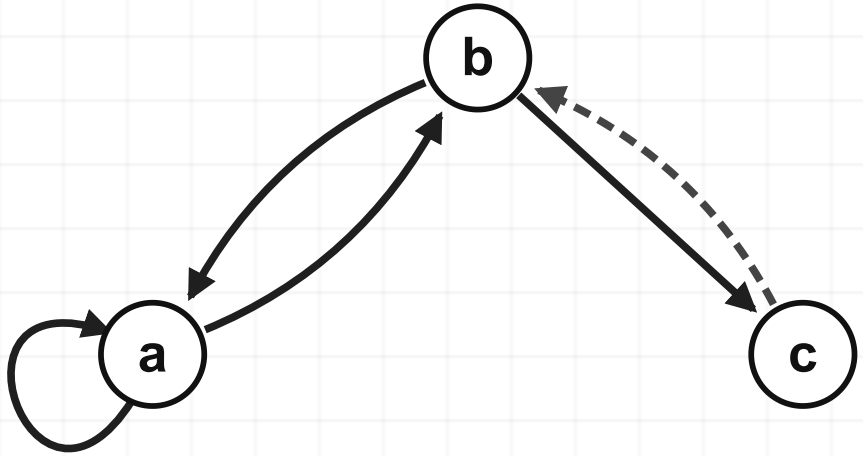


Every element of the underlying set needs a loop.

Symmetric closure: add reverse arrows

$$R = \{(a,a), (a,b), (b,a), (b,c)\}.$$

01 **Which pair must be added?**
(c,b), **the reverse of** (b,c).



An existing reverse arrow does not need to be added twice.

Two closures have direct formulas

Reflexive closure

$$r(R) = R \cup \Delta_A$$

$$\Delta_A = \{(a, a) : a \in A\}.$$

Symmetric closure

$$s(R) = R \cup R^{-1}$$

$$R^{-1} = \{(b, a) : (a, b) \in R\}.$$

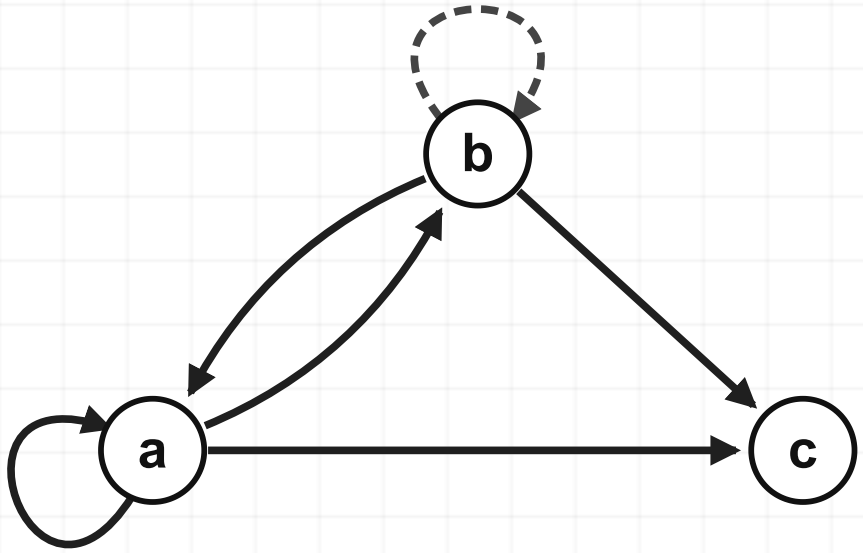
For each formula: containment, the required property, and minimality.

EXERCISE 4

A shortcut is not the whole repair

Add (a, c) because aRb and bRc. Is the result transitive?

01 **Which other pair is forced?**
(b, b)**, because** bRa **and** aRb**.**



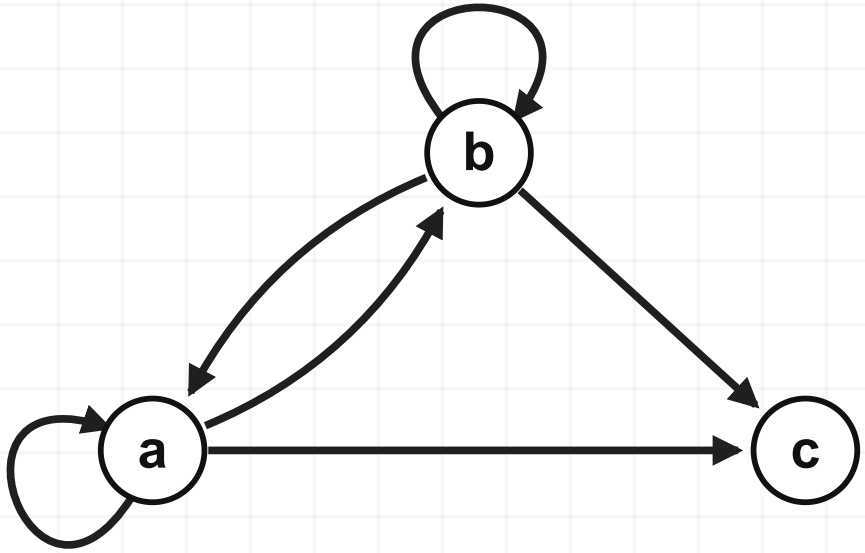
A two-edge path may begin and end at the same vertex.

The transitive closure is now complete

$$t(R) = R \cup \{(a, c), (b, b)\}.$$

From a: reach a, b, c. From b: reach a, b, c.

From c: no outgoing path, so no new pair.



Every two-step route already has its shortcut.

Transitive closure records reachability

$(x, y) \in t(R)$ exactly when there is a path from x to y using one or more edges of R.

> Contains R
>
> **An original edge is a path of length one.**

> Transitive
>
> **Concatenate a path from x to y with one from y to z.**

> Smallest
>
> **Every transitive relation containing R must contain every path shortcut.**

A loop is forced by a positive-length cycle, not by an empty path.

One pass may miss a longer route

$$A = \{1, 2, 3, 4\}, \quad R = \{(1, 2), (2, 3), (3, 4)\}.$$

01 **Find $t(R) \setminus R$.**

$\{(1, 3), (2, 4), (1, 4)\}$.

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02 **Why must (1,4) appear?**

It is forced by the path $1 \to 2 \to 3 \to 4$.

Continue until adding shortcuts creates no new obligations.

EXERCISE 6

When do symmetry and transitivity force loops?

Assume $\forall a \in A\ \exists b \in A :\ aRb \text{ or } bRa$.

01 **Must $t(s(R))$ be reflexive on A?**

Yes. Symmetry supplies both directions, then transitivity supplies each loop.

Use the incident edge in either direction

Fix an arbitrary element

Choose $c \in A$. There is $d \in A$ with cRd or dRc.

After symmetric closure

Both (c, d) and (d, c) belong to $s(R)$.

After transitive closure

$(c, c) \in t(s(R))$. This holds for every $c \in A$.

The same argument works when c = d.

EXERCISE 7

An isolated element changes the answer

$$A = \{a, b, c\}, \quad R = \{(a, b)\}.$$

01 **Find $t(s(R))$.**

$\{(a, a), (a, b), (b, a), (b, b)\}$.

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02 **Why is it not reflexive on A?**

(c, c) is missing: c has no incident edge.

Symmetric and transitive does not imply reflexive on the whole set.

Three repairs, three obligations

Closure	What must be added?	What may remain absent?
Reflexive	Every missing loop	Reverse arrows
Symmetric	Every missing reverse arrow	Loops at isolated elements
Transitive	Every path shortcut	Loops without a return path

Always name the underlying set before computing a closure.